

Rapid Cortex Data Flow & Architecture Overview

Plain-language technical overview for agency IT, procurement, and pilot stakeholders.

High-Level Architecture

- Marketing site and web app served through CloudFront/WAF
- Authenticated app access through Cognito
- Application APIs through AWS Lambda and/or ECS service routes
- DynamoDB for incident metadata, user state, audit records, and pilot settings
- S3 for controlled file/media storage
- Kinesis Video Streams or equivalent media service for live video sessions
- CloudWatch/CloudTrail for logging, monitoring, and operational evidence
- CAD integrations through read-only adapters during pilots

Caller Media Flow

Step	Description	Security Control
1	Dispatcher requests media support during a call	Role and agency authorization checked
2	System sends caller a secure SMS link	Link is short-lived and tied to incident/session
3	Caller uploads photo/video or starts live session	No public bucket access; media is encrypted
4	Dispatcher/supervisor views media in console	Access is logged with user, time, and incident
5	Retention policy expires or preserves content	Pilot retention rules enforced by agency configuration

Incident Intelligence Flow

Input	Processing	Output	Human Control
Call notes/transcript/media metadata	Assistive intelligence pipeline with validation	Summary, risks, language support, suggested context	Dispatcher and supervisor review before use
CAD read-only incident data	Adapter normalizes non-sensitive fields	Context panel inside Rapid Cortex	No write-back in pilot mode
Audit events	Append-only logging process	Agency audit report and compliance trail	Admin/export controls

Production Architecture Decision

The recommended production path is a separated dev/prod environment, CloudFront + WAF at the edge, ECS/Fargate or Amplify for the web app, smaller API stacks to avoid oversized SAM deployments, and automated health checks with rollback.